

SOURCE CODING

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Source Encoding is a process by which the output of an information Source is converted into a binary sequence as its output. The functional block that performs this task in the Communication system is called "Source Encoder".

The input to the source encoder is a symbol sequence emitted by information Source.

The Source encoder assigns variable length binary code words to block of symbols and produce a binary Sequence as its output.

Properties of codes

1. Block code : A block code is a code which maps each of the symbol of the source alphabet 'S' into some "finite Sequence" of code symbols from the code alphabet 'X' and each of these finite sequence is called a "code-word".

ex.: Consider a source S emitting only four symbols which are to be enclosed with binary coding

$$S = \{s_1, s_2, s_3, s_4\} \quad \text{and} \quad X = \{0, 1\}$$

A block code can be constructed by considering a pair of code symbols for each source symbol

Source Symbols	code-A
s_1	00
s_2	01
s_3	10
s_4	11

② Non-Singular : A block code is said to be "non-singular" if and only if all the code-words are distinct & easily "distinguishable" from one another.

ex:- Consider a Source emitting 4 symbols with $x = \{0, 1\}$
 i.e. $S = \{S_1, S_2, S_3, S_4\}$
 Let us assign code words for these source symbols

Source Symbols	code-B
S_1	0
S_2	00
S_3	01
S_4	11

The code-word given, appears to be non-singular, but actually it is not so, this can be ^{confirmed by} considering the second extension of these code-words which is shown below

Source Symbol	code B	Source Symbol	code B	S_1S_1	code B	S_1S_2	code B
S_1S_1	00	S_2S_1	000	S_3S_1	010	S_4S_1	100
S_1S_2	000	S_2S_2	0000	S_3S_2	0100	S_4S_2	1100
S_1S_3	001	S_2S_3	0001	S_3S_3	0101	S_4S_3	1101
S_1S_4	011	S_2S_4	0011	S_3S_4	0111	S_4S_4	1111

$\therefore$ The code-word for S_1S_2 and S_2S_1 are the "Same", thus we can conclude that the second extension of code-B are not "distinct" and hence they become "Singular"

③ Uniquely Decodable codes : A Non-Singular code is said to be uniquely Decodable code at the receiver, if every symbol in a long received sequence can be uniquely identified.

To understand this property let us consider second extension of "code A" which is uniquely Decodable code.

Source Symbols	code A	Source Symbols	code A	Source Symbols	code A	Source Symbols	code A
S_1S_1	0000	S_2S_1	0100	S_3S_1	1000	S_4S_1	1100
S_1S_2	0001	S_2S_2	0101	S_3S_2	1001	S_4S_2	1101
S_1S_3	0010	S_2S_3	0110	S_3S_3	1010	S_4S_3	1110
S_1S_4	0011	S_2S_4	0111	S_3S_4	1011	S_4S_4	1111

From the above table we can conclude that all codes are non-singular hence they can ^{be} uniquely decoded.

4. Instantaneous codes:- A uniquely Decodable codes is said to be "instantaneous codes" if it is possible to recognize the end of any code-word in any received sequence, without reference to the succeeding symbols.

EX:- Let us consider 3 uniquely decodable codes :-

Source Symbols	code-C	code-D	code-E
S_1	00	0	0
S_2	01	10	01
S_3	10	110	011
S_4	11	1100	0111

Let the received sequence be 001100

If code-C is used, then it is decoded as $S_1S_4S_1$.
 - " code-D _____ " _____ as $S_1S_1S_3S_1$.
 - " code-E _____ " _____ as $S_1S_3S_1S_1$.

When the code-C and D are used for Decoding the given sequence, the symbols arrive at the receiver are decoded without referring to the succeeding symbols.

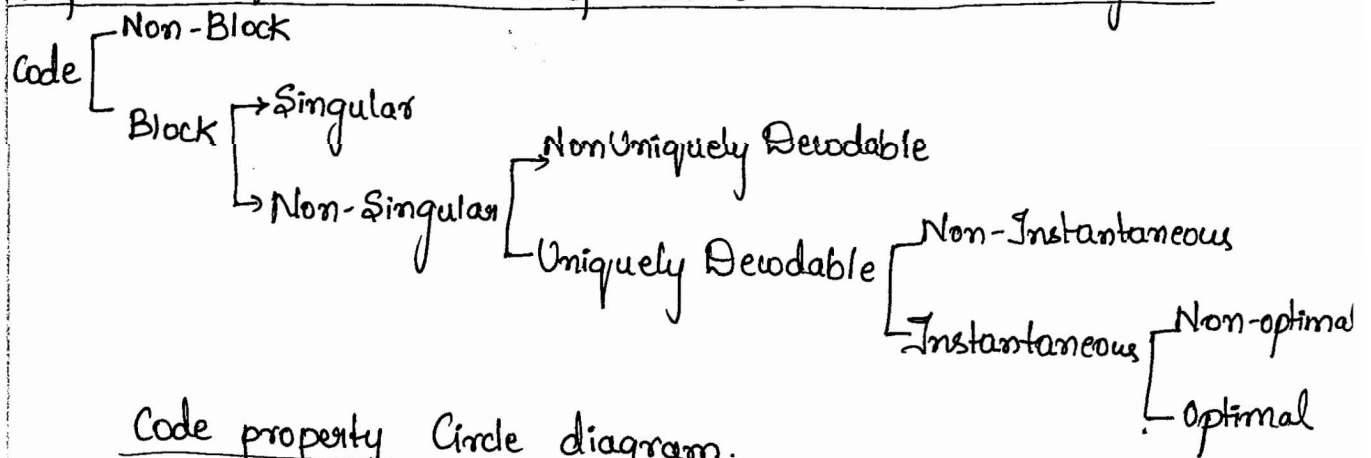
When code-E is used for decoding & when symbol '0' arrives at the receiver, we are not in a position to decode it. Since we have to wait for the succeeding symbols to arrive at the receiver. coz all the code-words in code-E start with '0'.

When the second symbol to arrives is '0', then the first '0' can be decoded as 'S', if the second symbol is '1' then again we have to wait for the succeeding symbols to arrive.

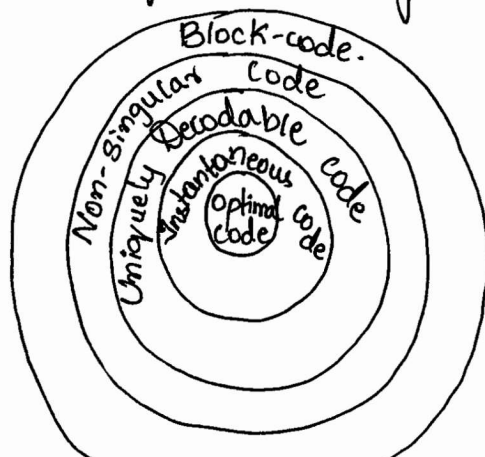
hence Code C, code D will be instantaneous codes
code E is not an instantaneous code.

⑥ Optimal Codes: An instantaneous code is said to be "optimal code" if it has "minimum average length L " for a source with a probability for the source symbol.

Properties of code is represented as a tree-diagram.



Code property Circle diagram.



There are many algorithms available for designing a source encoder. We have a simple most powerful source encoding algorithm given by Shannon.

Shannon's Encoding Algorithm

The design of the source encoder can be formulated as follows. The i/p to the source encoder consists of one of q possible messages, each message containing N symbols.

Let $P_1, P_2, P_3, \dots, P_q$ be the probabilities of these q messages. We would like to code the i th message of m_i using a unique binary code-word C_i of length n_i bits, where n_i is an integer.

Our objective is to find n_i and C_i for $i=1, 2, \dots, q$ such that the average number of bits/symbol " $\hat{H}_N$ " used in the coding scheme is as close to H_N as possible

$$\text{i.e. } \hat{H}_N = \frac{1}{N} \sum_{i=1}^q P_i n_i \longrightarrow \frac{1}{N} \sum_{i=1}^q P_i \log_2 \frac{1}{P_i}$$

Several solutions have been proposed to the above problem & the algorithm given by Shannon (& Fano) is stated below.

" Suppose the q messages $m_1, m_2, \dots, m_q$ are arranged in order of decreasing probabilities such that,

$$P_1 \geq P_2 \geq P_3 \dots \geq P_q$$

Let, $F_i = \sum_{k=1}^{i-1} P_k$ with $F_1 = 0$

Let, n_i be an integer such that

$$\log_2 \frac{1}{P_i} \leq n_i < 1 + \log_2 \frac{1}{P_i}$$

Then the code word for the message m_i is the binary expansion of the fraction f_i up to n_i bits

$$C_i = (f_i)_{\text{binary}} \text{ up to } n_i \text{ bits}$$

The algorithm yield a source coding procedure that has the following properties.

- ① Message of high probabilities are represented by short code words & those of low probabilities are represented by long code words.
- ② The code word for m_i will differ from all succeeding code words in one or more places & hence it is possible to decode messages uniquely from their code words.
- ③ The average number of bits/symbol used by the encoder is bounded by $G_N \leq \hat{H}_N < G_N + \frac{1}{N}$ where $N \rightarrow$ length of msg.
- ④ Rate efficiency e of encoder using block of 'N' symbols is defined as $e = \frac{H}{\hat{H}_N}$

Show that : $G_N \leq \hat{H}_N < G_N + \frac{1}{N}$

$$\rightarrow \text{if } \log_2\left(\frac{1}{P_i}\right) \leq n_i < 1 + \log_2\left(\frac{1}{P_i}\right) \longrightarrow \text{①}$$

$$\text{W.K.T } \hat{H}_N = \frac{1}{N} \sum_{i=1}^q P_i n_i, \quad G_N = \frac{1}{N} \sum_{i=1}^q P_i \log_2 \frac{1}{P_i}$$

Multiply equn ① by P_i/N

$$\frac{P_i}{N} \log_2 \frac{1}{P_i} \leq \frac{n_i P_i}{N} < \frac{P_i}{N} + \frac{P_i}{N} \log_2 \frac{1}{P_i}$$

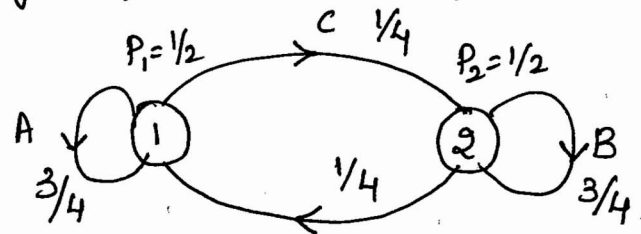
Taking $\sum_{i=1}^q$ on both sides

$$\frac{1}{N} \sum_{i=1}^q P_i \log_2 \frac{1}{P_i} \leq \frac{1}{N} \sum_{i=1}^q n_i P_i < \frac{1}{N} \left[\sum_{i=1}^q P_i + \sum_{i=1}^q P_i \log_2 \frac{1}{P_i} \right]$$

$$\Rightarrow \boxed{G_N \leq \hat{H}_N < \frac{1}{N} + G_N} \quad \because \sum_{i=1}^q P_i = 1$$

* Problems.

- Design a source encoder for the information source given below. Compare the average output bit rate and efficiency of the encoder for $n=1, 2$ and 3



$$\Rightarrow H = H_1 P_1 + H_2 P_2 + H_3 P_3$$

$$H = 0.8112 \text{ bits/symbol}$$

→ for $N=1$

m_i	P_i	n_i	f_i	C_i
A	$3/8$	2	0	00
B	$3/8$	2	$3/8$	01
C	$1/4$	2	$6/8$	11

To find f_i

We have $F_i = \sum_{k=1}^{i-1} P_k$ with $F_1 = 0$

$$\therefore F_1 = 0$$

$$F_2 = \sum_{k=1}^{2-1} P_k = F_2 = 3/8$$

$$F_3 = \sum_{k=1}^{3-1} P_k \Rightarrow \frac{3}{8} + \frac{3}{8} \Rightarrow F_3 = \frac{6}{8}$$

To find C_i

$C_1 = 00$ coz $f_1 = 0$ and length $n_i = 2 \therefore C_i = 00$

$C_2 = \frac{3}{8} = 0.375 \times 2 = 0.75 \Rightarrow 0$
 $0.75 \times 2 = 1.5 \Rightarrow 1$
 $1.5 \times 2 = 3.0 \Rightarrow 1$
 but we take $C_2 = 01$
 coz $n_i = 2$

To find n_i

$$\log_2 \frac{1}{P_i} \leq n_i < 1 + \log_2 \frac{1}{P_i}$$

$$\log_2 \frac{8}{3} \leq n_i < 1 + \log_2 \frac{8}{3}$$

$$1.4 \leq n_i < 2.4$$

$$n_i = 2$$

$\therefore$ as per formula n_i should be greater than 1.4 and it should be less than 2.4, so for coding purpose we have taken '2' where n_i is an integer

$$\hat{H}_1 = \frac{1}{\sum_{i=1}^3 P_i} \sum_{i=1}^3 P_i n_i$$

$$= \frac{3}{8} \times 2 + \frac{3}{8} \times 2 + \frac{1}{4} \times 2$$

$$\hat{H}_1 = 2 \text{ bits / symbol}$$

$$G_1 = \frac{1}{\sum_{i=1}^3 P_i} \sum_{i=1}^3 P_i \log_2 \frac{1}{P_i}$$

$$G_1 = \frac{3}{8} \log_2 \frac{8}{3} + \frac{3}{8} \log_2 \frac{8}{3} + \frac{1}{4} \log_2 4 \Rightarrow 1.5612 \text{ bits/symbol}$$

$$G_{N+1} = G_1 + \frac{1}{N} \Rightarrow 1.5612 + 1 = 2.5612$$

$$\text{Rate efficiency } e = \frac{H}{\hat{H}_N} \Rightarrow 40.56\%$$

for $N=2$

m_i	P_i	n_i	F_i	C_i
AA	$\frac{9}{32}$	2	0	00
BB	$\frac{9}{32}$	2	$\frac{9}{32}$	01
AC	$\frac{3}{32}$	4	$\frac{18}{32}$	1001
CB	$\frac{3}{32}$	4	$\frac{21}{32}$	1010
BC	$\frac{3}{32}$	4	$\frac{24}{32}$	1100
CA	$\frac{3}{32}$	4	$\frac{27}{32}$	1101
CC	$\frac{2}{32}$	4	$\frac{30}{32}$	1111

$$\hat{H}_2 = \frac{1}{\sum_{i=1}^7 P_i} \sum_{i=1}^7 P_i n_i$$

$$H_2 = \frac{1}{2} \left[\frac{9}{32} \times 2 \times 2 + \frac{3}{32} \times 4 \times 4 + \frac{2}{32} \times 4 \right]$$

$$H_2 = 1.4375$$

$$G_2 = \frac{1}{\sum_{i=1}^7 P_i} \sum_{i=1}^7 P_i \log_2 \frac{1}{P_i}$$

$$G_2 = \frac{1}{2} \left[\frac{9}{32} \log_2 \frac{32}{9} \times 2 + \frac{3}{32} \log_2 \frac{32}{3} \times 4 + \frac{2}{32} \log_2 \frac{32}{2} \right]$$

$$e = \frac{H}{\hat{H}_N} = \frac{1.4375}{2.5612} = 56.12\%$$

$$G_N + \frac{1}{N} \Rightarrow G_2 + \frac{1}{2} \Rightarrow 1.28 + \frac{1}{2} \Rightarrow 1.78$$

$$e = \frac{H}{\hat{H}_2} = \frac{0.8112}{1.4375}$$

$$e = 0.5643 \Rightarrow \boxed{e = 56.43\%}$$

for $N=3$

m_i	P_i	n_i	F_i	C_i
AAA	$\frac{27}{128}$	3	0	000
BBB	$\frac{27}{128}$	3	$\frac{27}{128}$	001
AAC	$\frac{9}{128}$	4	$\frac{54}{128}$	0110
ACB	$\frac{9}{128}$	4	$\frac{63}{128}$	0111
CBB	$\frac{9}{128}$	4	$\frac{72}{128}$	1001
CAA	$\frac{9}{128}$	4	$\frac{81}{128}$	1010
BCA	$\frac{9}{128}$	4	$\frac{90}{128}$	1011
BBC	$\frac{9}{128}$	4	$\frac{99}{128}$	1100
CBC	$\frac{3}{128}$	6	$\frac{108}{128}$	110110
CCA	$\frac{3}{128}$	6	$\frac{111}{128}$	110111
CAC	$\frac{3}{128}$	6	$\frac{114}{128}$	111001
CCB	$\frac{3}{128}$	6	$\frac{117}{128}$	111010
BCC	$\frac{3}{128}$	6	$\frac{120}{128}$	111100
ACC	$\frac{3}{128}$	6	$\frac{123}{128}$	111101
CCC	$\frac{2}{128}$	6	$\frac{126}{128}$	111111

$$\hat{H}_N = \frac{1}{N} \sum_{i=1}^q P_i n_i$$

$$\hat{H}_3 = \frac{1}{3} \sum_{i=1}^{15} P_i n_i$$

$$\hat{H}_3 = \frac{1}{3} \left[\frac{27}{128} \times 3 \times 2 + \frac{9}{128} \times 4 \times 6 + \frac{3}{128} \times 6 \times 6 + \frac{2}{128} \times 6 \right]$$

$$\hat{H}_3 = \frac{1}{3} [1.2656 + 1.6875 + 0.84375 + 0.09375]$$

$$\hat{H}_3 = \frac{1}{3} [3.8906] \Rightarrow 1.30$$

$$\boxed{\hat{H}_3 = 1.30 \text{ bits/symbol}}$$

$$G_N = \frac{1}{N} \sum_{i=1}^q P_i \log_2 \frac{1}{P_i}$$

$$G_3 = \frac{1}{3} \sum_{i=1}^{15} P_i \log_2 \frac{1}{P_i}$$

$$\boxed{G_3 = 1.139}$$

$$G_N \leq \hat{H}_N < G_N + \frac{1}{N}$$

$$1.13 \leq 1.3 < 1.47$$

$$e = \frac{H}{\hat{H}_3} = \frac{0.8112}{1.30} = 0.624$$

$$\boxed{e = 62.40\%}$$

∴ Performance of the encoder

Average number of bits per symbol used	N=1	N=2	N=3
$\hat{H}_N$	2	1.44	1.30
G_N	1.561	1.28	1.139
$G_N + \frac{1}{N}$	2.561	1.78	1.47
efficiency = $\frac{H}{\hat{H}_N}$	40.56%	56.43%	62.40%
$H = 0.8113$			

- ② A Source emits independent sequence of symbols from a Source alphabet containing 8 symbols $S_1, S_2, S_3, \dots, S_8$ with probabilities $\frac{1}{4}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}, \frac{1}{16}, \frac{1}{16}, \frac{1}{16}, \frac{1}{16}$. Design a Source encoder for a block size of 1.

Solⁿ

S_i	P_i	n_i	F_i	C_i
S_1	$\frac{1}{4}$	2	0	00
S_2	$\frac{1}{4}$	2	$\frac{1}{4}$	01
S_3	$\frac{1}{8}$	3	$\frac{2}{4}$	100
S_4	$\frac{1}{8}$	3	$\frac{5}{8}$	100
S_5	$\frac{1}{16}$	4	$\frac{6}{8}$	1100
S_6	$\frac{1}{16}$	4	$\frac{13}{16}$	1101
S_7	$\frac{1}{16}$	4	$\frac{14}{16}$	1110
S_8	$\frac{1}{16}$	4	$\frac{15}{16}$	1111

$$\rightarrow \log_2 \frac{1}{P_i} \leq n_i < \log_2 \frac{1}{P_i} + 1$$

$$(2) \leq n_i < 3$$

$$\rightarrow F_i = \sum_{k=1}^{i-1} P_k$$

$$\rightarrow \hat{H}_N = \frac{1}{N} \sum_{i=1}^q P_i n_i$$

$$\hat{H}_N = \frac{1}{1} \sum_{i=1}^8 P_i n_i$$

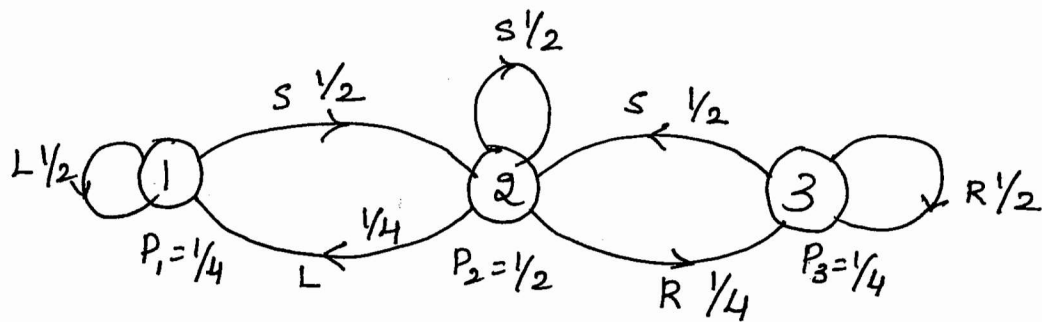
$$= \frac{1}{1} \left[\frac{1}{4} \times 2 \times 2 + \frac{1}{8} \times 3 \times 2 + \frac{1}{16} \times 4 \times 4 \right]$$

$$P = \frac{1}{N} \sum P_i n_i \quad \frac{1}{16} = 2 \text{ bits/symbol}$$

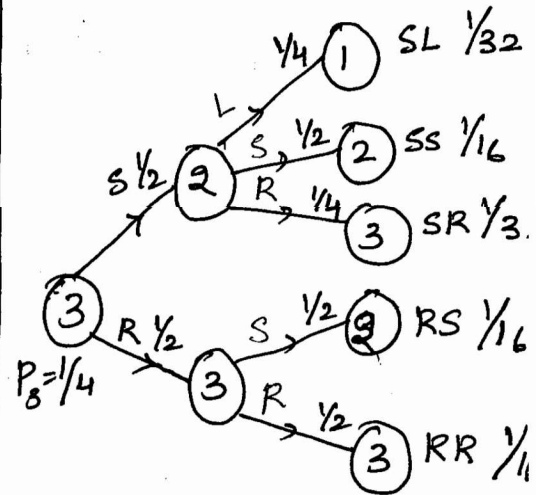
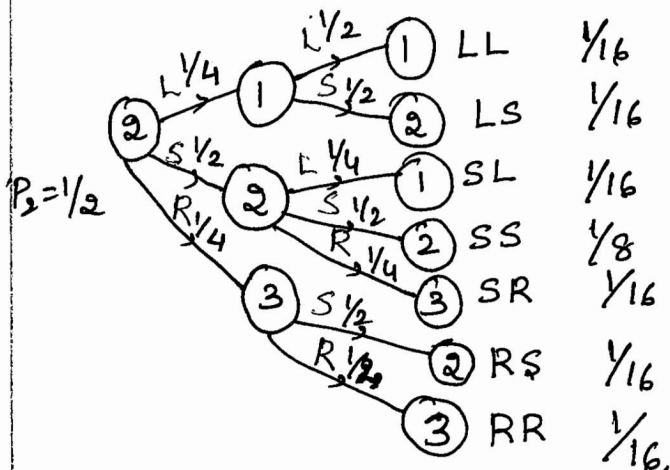
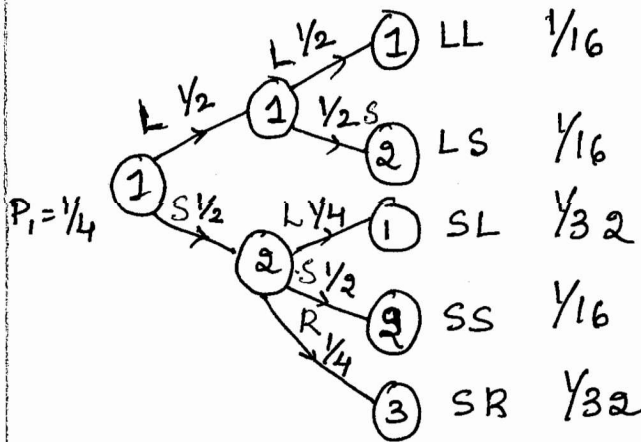
$\hat{H}$

1.30

- ③ For the given source design the source encoding scheme using a block size of 2 symbols & variable length code words. Calculate the actual number of bits/symbol used by the encoder and verify that $\hat{H}_2 \leq G_2 + 1/2$.
If the source is emitting symbols at a rate of 1000 symbols/sec. compute the output bit rate of the encoder



→ Given $N=2$.



m_i	P_i
LL	$1/8$
LS	$1/8$
SL	$1/8$
SS	$2/8$
SR	$1/8$
RS	$1/8$
RR	$1/8$

∴ So as per Source coding algorithm the probabilities should be arranged in decreasing order

m_i	P_i	n_i	f_i	C_i
SS	$2/8$	2	0	00
LS	$1/8$	3	$2/8$	010
SL	$1/8$	3	$3/8$	011
LL	$1/8$	3	$4/8$	100
SR	$1/8$	3	$5/8$	101
RS	$1/8$	3	$6/8$	110
RR	$1/8$	3	$7/8$	111

⇒ To calculate C_i

for SS ⇒ AS $f_1 = 0$ & $n_1 = 2$
 $C_i = 00$

LS ⇒ $F_2 = \frac{2}{8} \Rightarrow F_2 = 0.25$

∴ $0.25 \times 2 = 0.5 = 0$

$0.5 \times 2 = 1.0 = 1$

$0 \times 0 = 0 = 0$

SL ⇒ $F_3 = 3/8 \Rightarrow F_3 = 0.375$

$0.375 \times 2 = 0.75 = 0$

$0.75 \times 2 = 1.5 = 1$

$0.5 \times 2 = 1.0 = 1$

Similarly for F_4 to F_7

$$\therefore \hat{H}_N = \frac{1}{N} \sum_{i=1}^n P_i n_i$$

$$\hat{H}_2 = \frac{1}{2} \sum_{i=1}^7 P_i n_i \Rightarrow \frac{1}{2} \left[2 \times \frac{2}{8} + \frac{1}{8} \times 3 \times 6 \right]$$

$$\hat{H}_2 = \frac{1}{2} (2.75)$$

$$\hat{H}_2 = 1.375 \text{ bits/symbol}$$

$$G_N = \frac{1}{N} \sum_{i=1}^n P_i \log_2 \frac{1}{P_i} \text{ bits/symbol}$$

$$G_2 = \frac{1}{2} \left[\frac{2}{8} \log_2 \frac{8}{2} + \frac{1}{8} \log_2 8 \times 6 \right]$$

$$G_2 = 1.375 \text{ bits/symbol}$$

$$\hat{H}_2 \leq G_2 + \frac{1}{2}$$

$$\Rightarrow 1.375 \leq 1.875$$

$$H = 1(1/4) + 1.5(1/2) + \frac{1}{4}(1) = 1.25 \text{ bits/symbol}$$

$$R = r_s \cdot H \Rightarrow R = (1000) (1.25) \quad R = 1250 \text{ bits/sec}$$

Shannon-Fano encoding algorithm.

Shannon Fano encoding algorithm is another technique used in constructing a source encoder.

- A Source encoder consists of arranging the messages in decreasing order of its probability and dividing the messages into two almost equally probable groups.
- The message in the 1st group is given bit '0', the message in the 2nd group are given bit 1.
- The procedure is now applied for each group separately and continued until no further division is possible.
- Finally we get the code words for respective symbols.

Problems.

- ① Find the code words occurring in the probability $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}$ & $\frac{1}{8}$ for symbols S_1, S_2, S_3, S_4

→

Symbols	P_i	I	II	III	C_i	length n_i
S_1	$\frac{1}{2}$	0			0	1
S_2	$\frac{1}{4}$	1	$\frac{1}{4}$	10		2
S_3	$\frac{1}{8}$		$\frac{1}{8}$	110		3
S_4	$\frac{1}{8}$		$\frac{1}{8}$	111		3

$$H = \frac{1}{2} \log 2 + \frac{1}{4} \log 4 + \frac{1}{8} \log 8 \times 2 \Rightarrow H = 1.75 \text{ bits/sy}$$

$$\hat{H}_1 = \frac{1}{1} \sum_{i=1}^4 P_i n_i \Rightarrow \hat{H}_1 = 1.75 \text{ bits/symbol}$$

$$\therefore e = \frac{H}{\hat{H}} \Rightarrow \frac{1.75}{1.75} = 100\%$$

2. Find the code word for the probability
 $\frac{1}{4}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}, \frac{1}{16}, \frac{1}{16}, \frac{1}{16}, \frac{1}{16}$ for the symbols
 $S_1, \dots, S_8$.

→ Symbols	P_i					ci	n_i
S_1	$\frac{1}{4}$	0	$\frac{1}{4}$	00		00	2
S_2	$\frac{1}{4}$		$\frac{1}{4}$	01		01	2
S_3	$\frac{1}{8}$	10	$\frac{1}{8}$	$\frac{1}{8}$ 100		100	3
S_4	$\frac{1}{8}$		$\frac{1}{8}$	$\frac{1}{8}$ 101		101	3
S_5	$\frac{1}{16}$	11	$\frac{1}{16}$	$\frac{1}{16}$ 110	$\frac{1}{16}$ 1100	1100	4
S_6	$\frac{1}{16}$		$\frac{1}{16}$		$\frac{1}{16}$ 1101	1101	4
S_7	$\frac{1}{16}$		$\frac{1}{16}$	$\frac{1}{16}$ 111	$\frac{1}{16}$ 1110	1110	4
S_8	$\frac{1}{16}$		$\frac{1}{16}$		$\frac{1}{16}$ 1111	1111	4

$$H = \left[\frac{1}{4} \log_2 4 \right] \times 2 + \left[\frac{1}{8} \log_2 8 \right] \times 2 + \left[\frac{1}{16} \log_2 16 \right] \times 4$$

$$H = 2.75 \text{ bits/symbol}$$

$$\hat{H}_1 = \frac{1}{1} \left[\frac{1}{4} \times 2 \times 2 + \frac{1}{8} \times 3 \times 2 + \frac{1}{16} \times 4 \times 4 \right] \quad \hat{H}_1 = 2.75 \text{ bits/symbol}$$

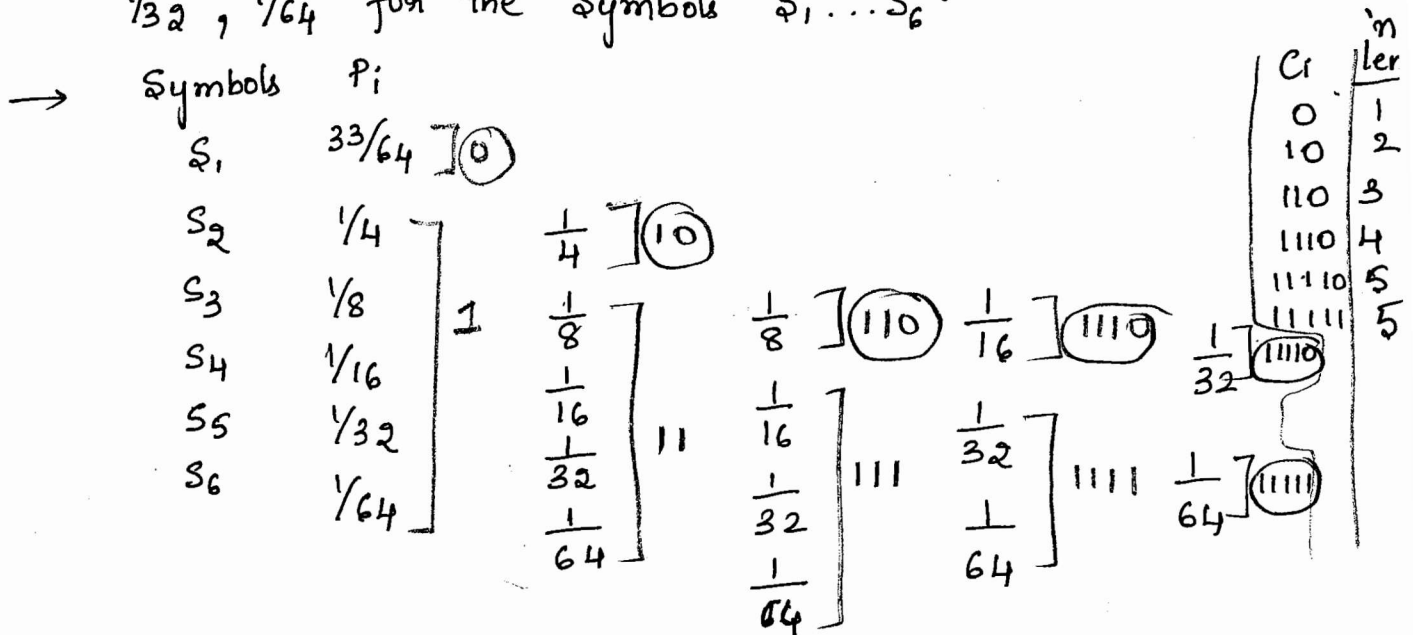
$$e = \frac{H}{\hat{H}} = \frac{2.75}{2.75} = 100\% \Rightarrow \boxed{e \leq 100\%} \text{ Note}$$

Redundancy: The Coding Redundancy denoted by R_c is defined as

$$R_c = 1 - e \Rightarrow 1 - 100\% = 0\%$$

where efficiency and Redundancy are expressed as percentages.

- Using Shannon Fano encoding algorithm,
 (3) Find the code word for the probability $\frac{33}{64}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{64}$ for the symbols $S_1, \dots, S_6$



$$H = \frac{33}{64} \log_2 \frac{64}{33} + \frac{1}{4} \log_2 4 + \frac{1}{8} \log_2 8 + \frac{1}{16} \log_2 16 + \frac{1}{32} \log_2 32 + \frac{1}{64} \log_2 64$$

$$H = 1.867 \text{ bits/symbol}$$

$$\hat{H}_N = \frac{1}{N} \sum_{i=1}^n P_i \cdot n_i$$

$$\hat{H}_1 = \frac{1}{1} \sum_{i=1}^6 n_i P_i$$

$$\hat{H}_1 = \frac{33}{64} \times 1 + \frac{1}{4} \times 2 + \frac{1}{8} \times 3 + \frac{1}{16} \times 4 + \frac{1}{32} \times 5 + \frac{1}{64} \times 5$$

$$\hat{H}_1 = 1.875 \text{ bits/symbol}$$

(4)

Symbols	P_i		
S_1	$\frac{1}{3}$	] 0	$\frac{1}{3}$] 00
S_2	$\frac{1}{3}$		$\frac{1}{3}$] 01
S_3	$\frac{1}{6}$	] 1	$\frac{1}{6}$] 10
S_4	$\frac{1}{6}$		$\frac{1}{6}$] 11
S_5	$\frac{1}{24}$	] 11	$\frac{1}{24}$
S_6	$\frac{1}{24}$		$\frac{1}{24}$

2nd Method.

Another way of generating code word for message consists of arranging the messages in decreasing order of probability and assigning the code words as follows.

The code word for 1st message is 0,

the code word for i^{th} message consists of $i-1$ bits of ones followed by a zero.

The code word for the last message is all ones. Such that the no of bits in the code word for the last message is equal to the total no of messages that have to be encoded.

Problem.

- a) find the code words & the avg number of bits/message used if the source emits one of 5 messages with probabilities $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$ and $\frac{1}{16}$.

$\rightarrow$	m_i	P_i	c_i	n_i
	m_1	$\frac{1}{2}$	0	1
	m_2	$\frac{1}{4}$	10	2
	m_3	$\frac{1}{8}$	110	3
	m_4	$\frac{1}{16}$	1110	4
	m_5	$\frac{1}{16}$	11111	5

$$\therefore \hat{H}_N = 1.9375 \text{ bits/symbol}$$

$$H = 1.875 \text{ bits/symbol}$$

$$e = \frac{H}{\hat{H}_N}$$

$$e = 96.77\%$$

- ② A Source produces 2 symbols with probabilities $\frac{3}{4}$ and $\frac{1}{4}$. Design a variable length code using Shannon-Fano algorithm, so that the coding efficiency is atleast 85%.

→

S_i	P_i	n_i
S_1	$\frac{3}{4}$	0
S_2	$\frac{1}{4}$	1

$$H_1 = \sum_{i=1}^2 P_i \log_2 \frac{1}{P_i}$$

$$H = \frac{3}{4} \log_2 \frac{4}{3} + \frac{1}{4} \log_2 4$$

$$H = 0.8112 \text{ bits/symbol}$$

$$\hat{H}_N = \frac{1}{N} \sum_{i=1}^N P_i n_i \quad \text{where } N=1$$

$$\hat{H}_N = \frac{1}{1} \times \left[\frac{3}{4} \times 1 + \frac{1}{4} \times 1 \right]$$

$$\hat{H}_N = 1 \text{ bit/symbol}$$

$$e = \frac{H}{\hat{H}_N} = \frac{0.8112}{1} = 81.12\%$$

As they have asked to find the coding efficiency for atleast 85%, we will go for length i.e. $N=2$

S_i	P_i	I	II	III	n_i
$S_1 S_1$	$\frac{9}{16}$	0			1
$S_1 S_2$	$\frac{3}{16}$	$\frac{3}{16}$] 10			2
$S_2 S_1$	$\frac{3}{16}$	$\frac{3}{16}$] 11			3
$S_2 S_2$	$\frac{1}{16}$	$\frac{3}{16}$] 111			3

$$\hat{H}_2 = \frac{1}{2} \sum_{i=1}^4 P_i n_i = \frac{1}{2} [$$

$$] = 0.8437$$

$$e = \frac{H}{\hat{H}_2} = 96.15\%$$

HUFFMAN CODING.

The huffman encoding algorithm is

- ① Arrange the symbols/messages in the decreasing order of probability.

- ② Check the relation, $n = r + \alpha(r-1)$ is satisfied.

where, $n = \text{no of symbols}$

$r = \text{length of the code alphabet}$

if $r = 2$ ($x = 0, 1$) $\rightarrow$ Binary huffman coding.

$r = 3$ ($x = 0, 1, 2$) $\rightarrow$ Ternary huffman coding.

$r = 4$ ($x = 0, 1, 2, 3$) $\rightarrow$ Quaternary huffman coding.

$\alpha = \text{no of reduction stages. } \alpha \text{ should be integer.}$

if the above relation doesnot give ' α ' as integer, add a dummy symbol with probability equal to '0' so that the above relation is satisfied with α being an integer.

- ③ Combine last ' r ' symbols and move the combined symbol as high as possible.
- ④ Repeat the above step till exactly ' n ' symbols are left in last stage.
- ⑤ the code for each symbol is found by moving backward to forward direction.

- ⑥ Calculate efficiency by using formula
$$e = \frac{H}{\bar{L} \log_2 r}$$

where $\bar{L}$ is Average code word length $\bar{L} = \sum_{i=1}^n P_i n_i$

- ⑦ Calculate variance using
$$\sigma^2 = \sum_{i=1}^n P_i (n_i - \bar{L})^2$$

Problems.

- ① A discrete memoryless source has a alphabet of 5 symbols with their probabilities given below. find huffman codes for the source and also find the efficiency and variance.

S_i	P_i	I	II	III	code	n_i
S_1	0.4	0.4	0.4	0.6	00	2
S_2	0.2	0.2	0.4	0.4	10	2
S_3	0.2	0.2	0.2	0.2	11	2
S_4	0.1	0.2	0.2	0.1	010	3
S_5	0.1	0.1	0.1	0.1	011	3

$$n = 2 + 2(r-1)$$

$$5 = 2 + 2(2-1)$$

$$2 = 5 - 2 \Rightarrow \boxed{2 = 3}$$

$$n = 2 + 2(r-1)$$

$$5 = 2 + 2(2-1)$$

$$2 = 5 - 2 \Rightarrow \boxed{2 = 3}$$

$$H = \sum_{i=1}^5 P_i \log_2 \frac{1}{P_i} \Rightarrow H = 0.4 \log_2 \frac{1}{0.4} + 0.2 \log_2 \frac{1}{0.2} \times 2 + 0.1 \log_2 \frac{1}{0.1} \times 3$$

$$\boxed{H = 2.1216 \text{ bits/symbol}}$$

$$\bar{L} = \sum_{i=1}^5 P_i n_i \Rightarrow 0.4 \times 2 + 0.2 \times 2 \times 2 + 0.1 \times 3 \times 2$$

$$\boxed{\bar{L} = 2.2 \text{ bits/symbol}}$$

$$e = \frac{H}{\bar{L} \log_2 2} \Rightarrow \frac{2.1216}{2.2 \log_2 2} \Rightarrow \boxed{e = 96.4\%}$$

$$\sigma^2 = \sum_{i=1}^5 P_i (n_i - \bar{L})^2 = 0.4(2 - 2.2)^2 + 0.2(2 - 2.2)^2 \times 2 + 0.1(3 - 2.2)^2$$

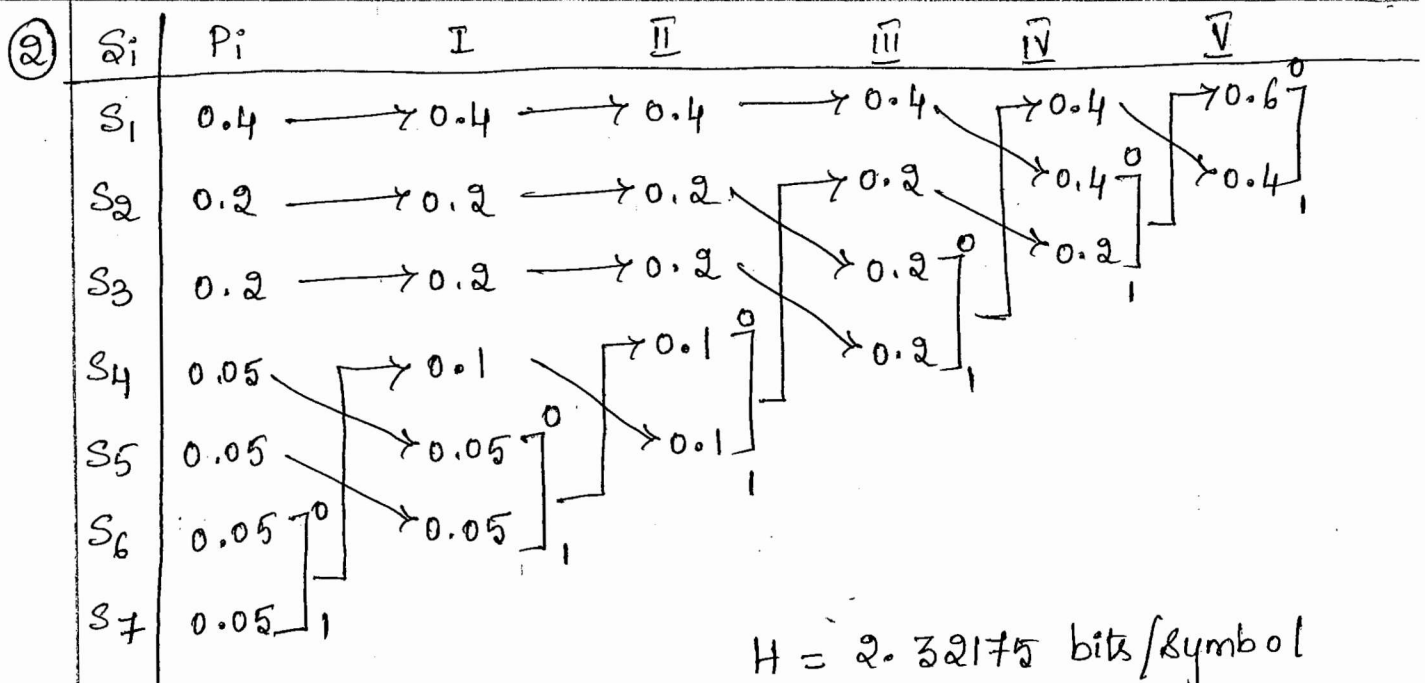
$$\boxed{\sigma = 0.16}$$

The variance should be as low as possible.

• Huffman coding as high as possible is preferable.

Note: Variance is a measure of code word length. we prefer as high as possible. Since Variance should be as low as possible.

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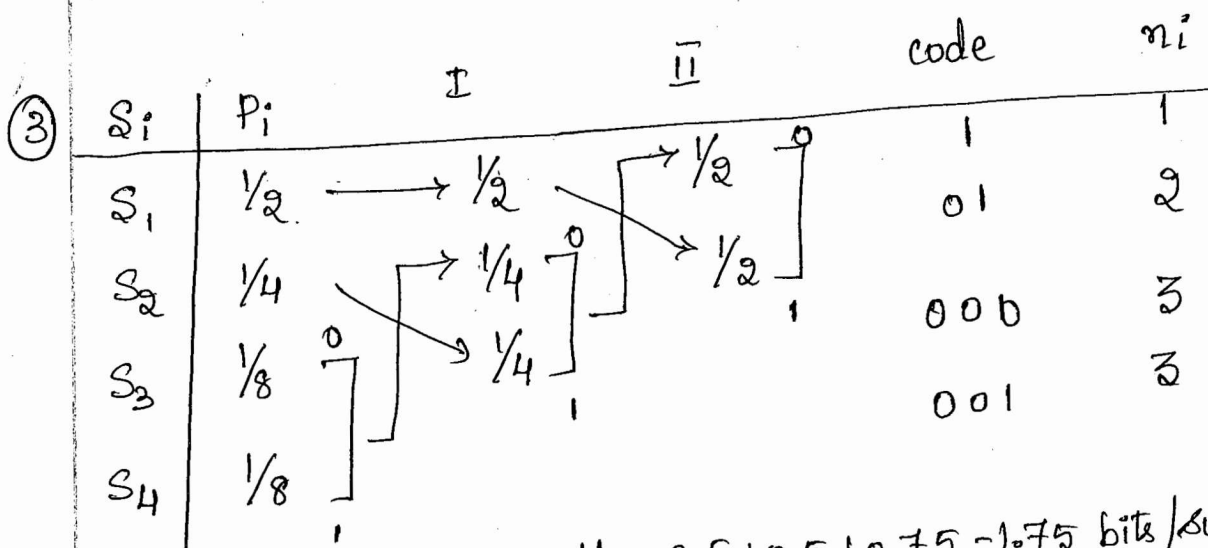


$$H = 2.32175 \text{ bits/symbol}$$

$$\bar{L} = \sum_{i=1}^7 P_i n_i = 2.4 \text{ bits/symbol}$$

$$e = \frac{H}{\bar{L}} = 96.7 \%$$

codes	n_i
00	2
10	2
11	2
0100	4
0101	4
0110	4
0111	4



$$H = 0.5 + 0.5 + 0.75 = 1.75 \text{ bits/symbol}$$

$$\bar{L} = \sum_{i=1}^3 P_i n_i = 1.75 \text{ bits/symbol}$$

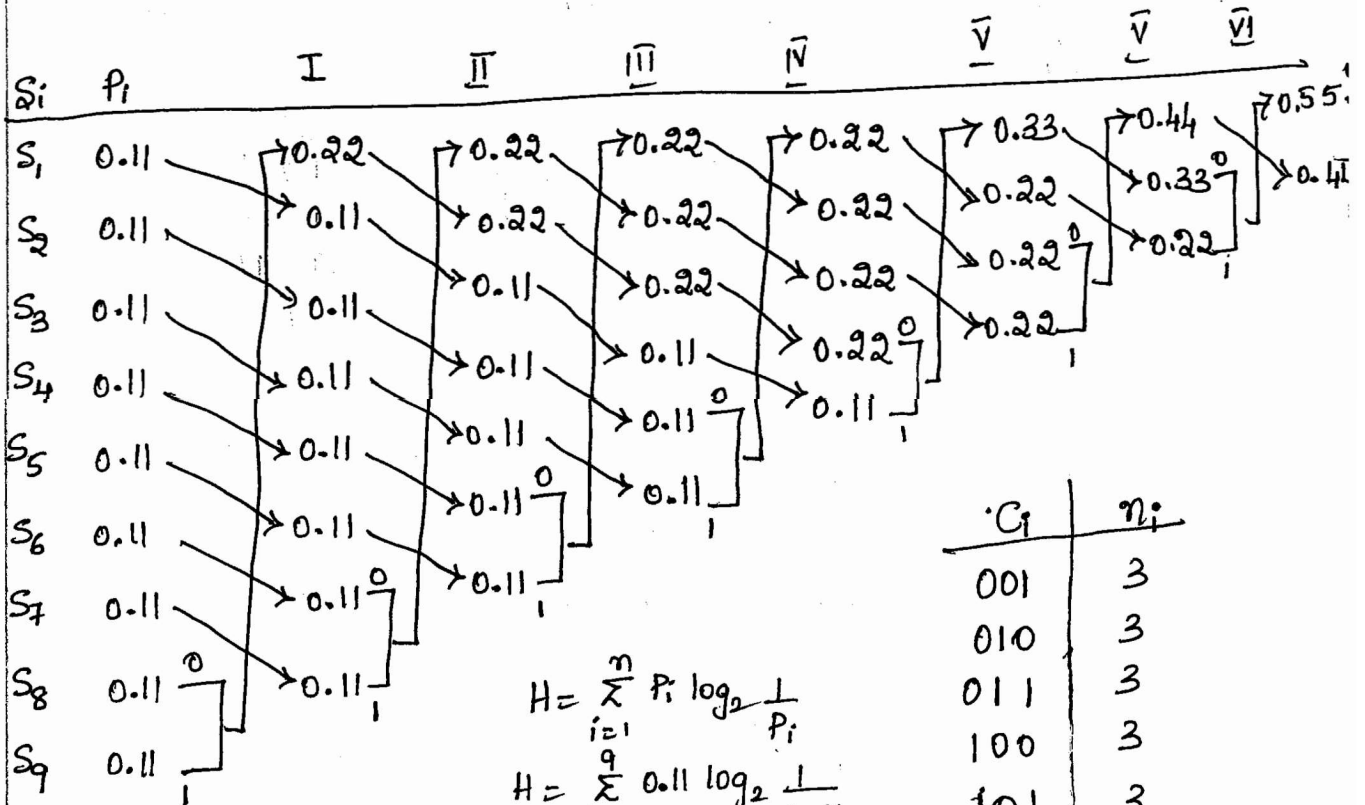
$$e = \frac{H}{\bar{L}} = \frac{1.75}{1.75} = 100 \%$$

④ A Source produces 9 symbols, that are equiprobable

- (a) Construct binary huffman coding by moving the Combined symbol as high as possible. Also find efficiency and variance of coding.
- (b) Construct binary huffman code by moving the Combined symbol as low as possible. Also find efficiency and variance of coding.

9 Symbol $\rightarrow 0.11, 0.11, 0.11, 0.11, 0.11, 0.11, 0.11, 0.11, 0.11$

→ ① As high as possible $\Rightarrow$ $n = r + \alpha(r-1)$
 $9 = 2 + \alpha(2-1) \Rightarrow \boxed{\alpha = 7}$



$$e = \frac{H}{L \log_2 8} = \frac{3.152}{3.19 \cdot \log_2 2}$$

$$e = 98.8\%$$

$$\sigma^2 = \sum_{i=1}^n p_i (x_i - \bar{x})^2$$

$$\sigma^2 = 0.11(3-3.19)^2 \times 7 + 0.11(6-2.10)^2 \times 9 \rightarrow \sigma^2 = 0.179 \Rightarrow \sigma = 0.414$$

$$H = \sum_{i=1}^n p_i \log_2 \frac{1}{p_i}$$

$$H = \sum_{|z|=1}^9 0.11 \log_2 \frac{1}{0.11}$$

$$H = 0.11 \log_2 \frac{1}{0.11} \times 9$$

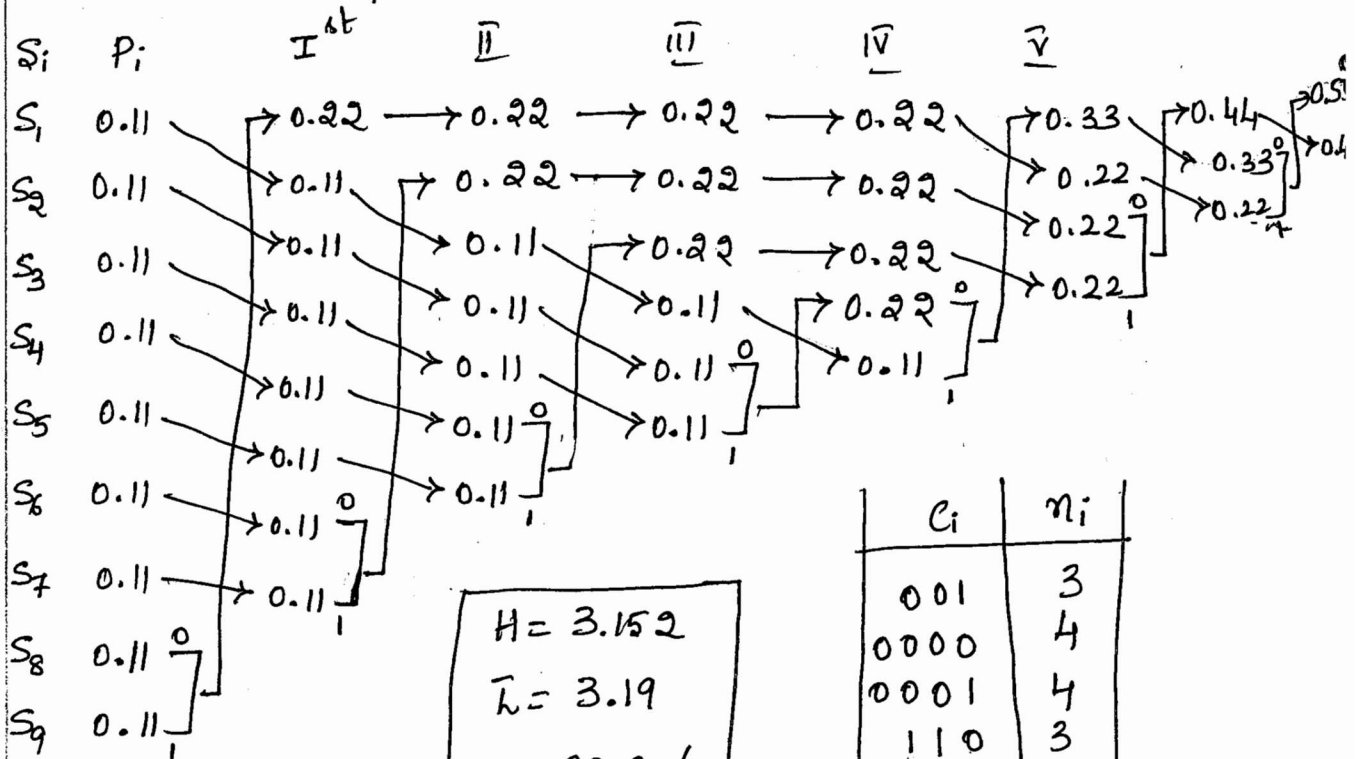
$$H = 3.152$$

$$\bar{L} = \sum_{i=1}^n p_i \cdot n_i$$

$$\bar{L} = \sum_{j=1}^9 p_j \eta_j \quad \boxed{\bar{L} = 3.19}$$

$\cdot C_i$	n_i
001	3
010	3
011	3
100	3
101	3
110	3
111	3
0000	4
0001	4

② As low as possible.



$$H = 3.152$$

$$\bar{h} = 3.19$$

$e = 98.8\%$

$$\sigma = 0.414$$

c_i	n_i
001	3
0000	4
0001	4
110	3
011	3
100	3
011	3
010	3
110	3

③ A Source produces 9 symbols, Construct ternary, Quaternary, Huffman coding by moving symbols as high as possible. Also find efficiency & variance of coding.

$$P_i = 0.11, 0.11, 0.11, 0.11, 0.11, 0.11, 0.11, 0.11, 0.11$$

→ So Given: $n = 9$ symbols

$r=3$ ($\because$ ternary huffman coding $r=0,1,2$)

$$\alpha = 9$$

$$\therefore n = r + 2(r-1)$$

$$q = 3 + 2(3-1) \Rightarrow \boxed{\gamma = 3}$$

S_i	P_i	1	2	3	0	C_i	n_i
S_1	0.11	→ 0.33	→ 0.33	→ 0.33	1	00	2
S_2	0.11	→ 0.11	→ 0.33	→ 0.33	2	01	2
S_3	0.11	→ 0.11	→ 0.11	0		02	2
S_4	0.11	→ 0.11	→ 0.11	1		10	2
S_5	0.11	→ 0.11	→ 0.11	2		11	2
S_6	0.11	→ 0.11	0			12	2
S_7	0.11	0				20	2
S_8	0.11	1				21	2
S_9	0.11	2				22	2

$$H = 3.169 \text{ bits/symbol}$$

$$\bar{L} = 1.98 \text{ bits/symbol}$$

$$e = 100\%$$

$$\sigma = 0.019$$

Quaternary Coding

$$n = r + d(r-1)$$

$$9 = 4 + 3d \quad \boxed{d = 1.6}$$

d should be integer. Add dummy symbol with prob = 0.

$$10 = 4 + 3d \Rightarrow \boxed{d = 2}$$

S_i	P_i	1st	2nd	C_i	n_i
S_1	0.11	→ 0.33	→ 0.44	2	1
S_2	0.11	→ 0.11	→ 0.33	3	1
S_3	0.11	→ 0.11	→ 0.11	00	2
S_4	0.11	→ 0.11	0	01	2
S_5	0.11	→ 0.11	1	02	2
S_6	0.11	→ 0.11	2	03	2
S_7	0.11	0		10	2
S_8	0.11	1		11	2
S_9	0.11	2		12	2
S_{10}	0.03	3		13	2

$$H = 3.169$$

$$\bar{L} = 1.76$$

$$e = 89.51\%$$

$$\sigma =$$

- ④ Construct Quaternary Huffman Coding by moving symbols as high as possible. find efficiency and variance.
 $S_1 = 1/3, S_2 = 1/4, S_3 = 1/8, S_4 = 1/8, S_5 = 1/12, S_6 = 1/12$

→ $n = 6$ symbols
 $r = 4$ (Quaternary)
 $\alpha = ?$

$$n = r + \alpha(r-1)$$

$$6 = 4 + \alpha(4-1)$$

$$6 - 4 = 3\alpha$$

$$\alpha = 0.667$$

α should be integer add dummy variable with probability equal to '0'

$$n = 7 \Rightarrow \boxed{n = 6 + 1}$$

$$\Rightarrow 7 - 4 = 3\alpha$$

$$\boxed{\alpha = 1}$$

S_i	P_i	1 st stage	C_i	n_i
S_1	$1/3$	$1/3 (0.33)$	0	1
S_2	$1/4$	$7/24 (0.29)$	2	1
S_3	$1/8$	$1/4 (0.25)$	3	1
S_4	$1/8$	$1/8 (0.125)$	10	2
S_5	$1/12$		11	2
S_6	$1/12$		12	2
S_7	0		13	2

$$\boxed{H = 2.3754 \text{ bits/symbol}}$$

$$\boxed{L = 1.292 \text{ bits/symbol}}$$

$$e = \frac{H}{L \log_2 4}$$

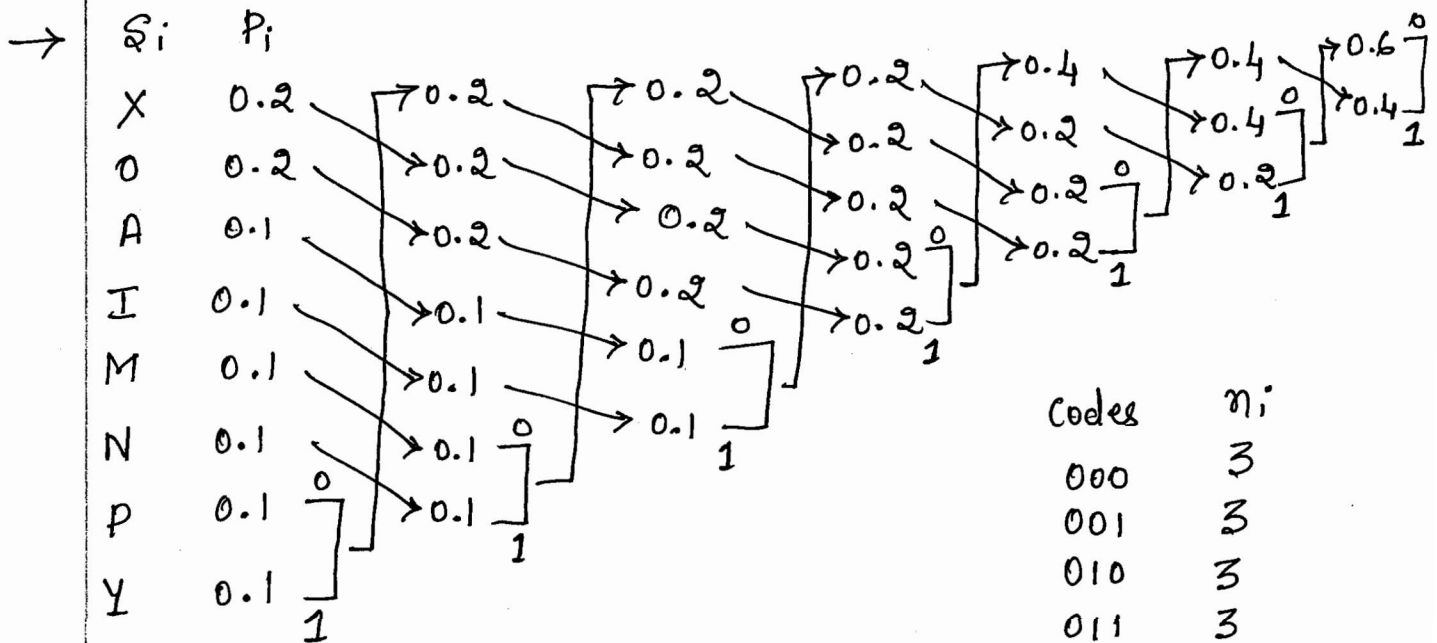
$$\boxed{e = 91.96\%}$$

$$\boxed{\sigma = 0.2065}$$

- ⑤ Consider a sequence of letters from english alphabet with their probability of occurrence as

A	I	X	M	N	O	P	Y
0.1	0.1	0.2	0.1	0.1	0.2	0.1	0.1

Compute 2 different huffman codes for this alphabet. for the two codes find average code word length & variance of code word length.



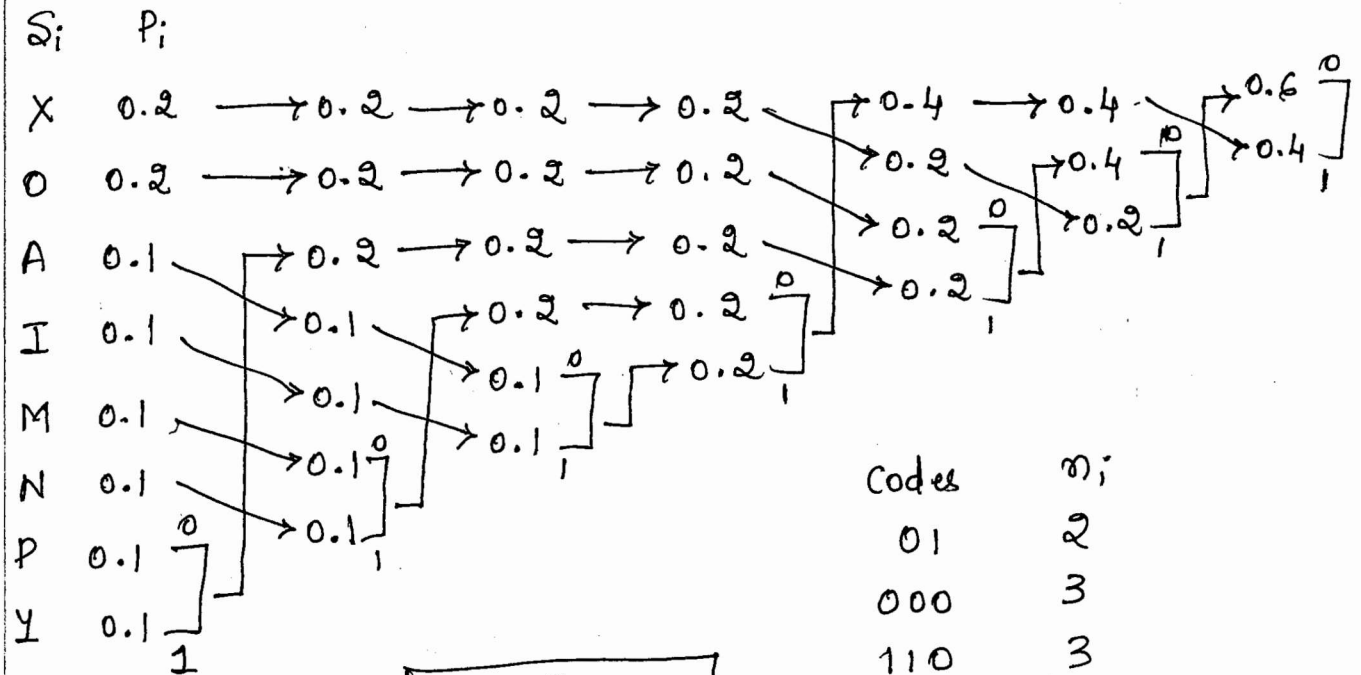
$$H = 2.9218$$

$$\bar{L} = 3 \text{ bits/symbol}$$

$$\sigma^2 = \sum_{i=1}^n p_i (n_i - \bar{L})^2$$

$$= 0.2 (3-3)^2 \times 2 + 0.1 (3-3)^2 \times 6$$

$$\sigma^2 = 0$$



$$H = 2.9218$$

$$\bar{L} = 3$$

$$O^2 = 0.4$$

Codes	n_i
01	2
000	3
110	3
111	3
100	3
101	3
0010	
0010	4
0011	4

So we prefer as high as possible.